L2.6a Optimization

§3.7 — optimization equation, constraint, interval

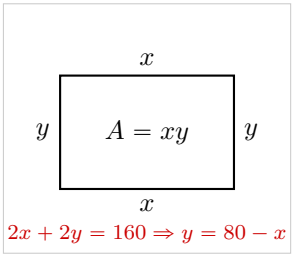
The optimization template

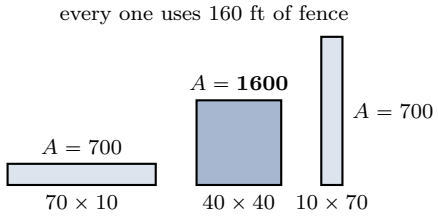
①	**Picture**	Draw it in a box and label every length.
②	**Sort the question**	**OEq** (optimization equation): the equation of the quantity to Maximize or minimize **Constraint**: the equation the variables must obey **Interval**: the values the picture and the question allow **Q?** Max/min OEq on $[a, b]$ subject to constraint
③	**Find critical pts**	Use the constraint to eliminate a variable, differentiate, and solve $f' = 0$ or f' undefined.
④	**Does this Maximize/min?**	2nd Derivative Test: $f'' < 0$ is a Max, $f'' > 0$ is a min.
⑤	**Check endpoints** Closed Interval Method	Evaluate at every critical number and both ends, compare, then back-substitute to answer the question asked.

Closed Interval Method. If f is continuous on $[a, b]$, its absolute Max and min are among $f(a)$, $f(b)$ and $f(c)$ at every critical number c. Evaluate them all and compare.

Rectangle of maximum area

Ex. 1 Find the rectangle with perimeter 160 ft whose area is as large as possible.

① 



② **OEq** $A = xy$ **Constraint** $2x + 2y = 160$

Interval $x \in [0, 80]$

Q? Max A on $[0, 80]$ subject to $2x + 2y = 160$

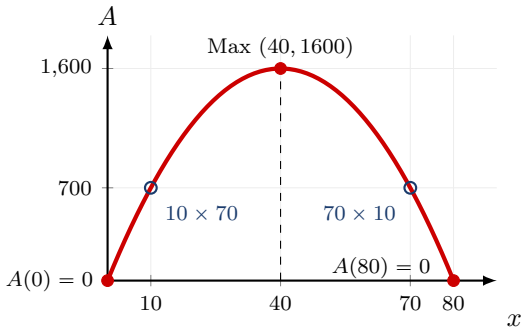
③ $2x + 2y = 160 \Rightarrow y = 80 - x$

$A = x\,y = x\,(80 - x) = 80x - x^2$

$A' = 80 - 2x = 0$, so $x = 40$

④ $A'' = -2$: CD, so a Max

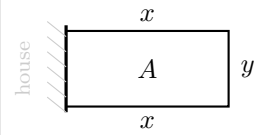
⑤ $A(0) = A(80) = 0$, $A(40) = \mathbf{1600}$



40 ft by 40 ft, $A = 1600$ ft^2

Optimization on a restricted interval

Ex. 2 Sarah builds a rectangular fence next to her house, using the house as one whole side. She has 40 ft of fencing, and no side of the plot may be shorter than 5 ft. What is the largest area she can enclose, and what is the smallest?

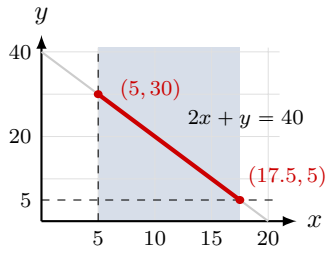


OEq $A = xy$

Constraint $2x + y = 40$ (the house side needs no fence)

eliminate y $y = 40 - 2x$, so $A = x(40 - 2x) = 40x - 2x^2$

Interval. No side shorter than 5 ft means $x \geqslant 5$ and $y \geqslant 5$, which cut off both ends of the constraint line:



$x \geqslant 5$ left end: $x = 5$

$y \geqslant 5$ $40 - 2x \geqslant 5$, so $35 \geqslant 2x$, so $x \leqslant 17.5$

$$\boxed{x \in [5,\ 17.5]}$$

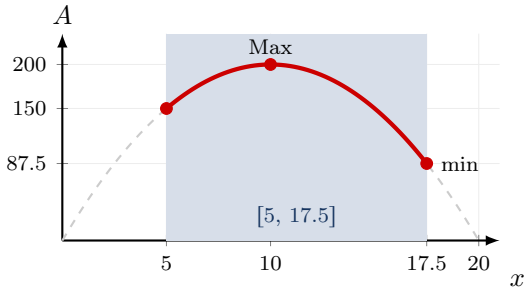
Q? Max and min A on $[5, 17.5]$ subject to $2x + y = 40$

Critical number. $A' = 40 - 4x = 0$, so $x = 10$. $A'' = -4$, so A is CD everywhere and $x = 10$ is a Max.

Closed Interval Method. A is continuous on $[5, 17.5]$, so check the critical number and both ends:

x	$A(x)$	
5 (left end)	150	
10 (critical)	**200**	**Max**
17.5 (right end)	**87.5**	**min**

$A'(17.5) = -30 \neq 0$, so no critical number gives the min; the endpoint does.



Back-substitute. $x = 10$ gives $y = 40 - 2(10) = 20$; $x = 17.5$ gives $y = 40 - 35 = 5$.

$$\boxed{\text{Max } A = 200 \text{ ft}^2 \text{ @ } 10 \text{ ft by } 20 \text{ ft}} \qquad \boxed{\text{min } A = 87.5 \text{ ft}^2 \text{ @ } 17.5 \text{ ft by } 5 \text{ ft}}$$